

Multiple Choice and True/False

1. This is a collection of programming statements that specify the fields and methods that a particular type of object may have.

- a. class
- b. method
- c. parameter
- d. instance

2. A class is analogous to a(n) _____.

- a. house
- b. blueprint
- c. drafting table
- d. architect

3. An object is a(n) _____.

- a. blueprint
- b. primitive data type
- c. variable
- d. instance of a class

4. This is a class member that holds data.

- a. method
- b. instance
- c. field
- d. constructor

5. This key word causes an object to be created in memory.

- a. create
- b. new
- c. object
- d. construct

6. This is a method that gets a value from a class's field, but does not change it.

- a. accessor
- b. constructor
- c. void
- d. mutator

7. This is a method that stores a value in a field or in some other way changes the value of a field.

- a. accessor
- b. constructor
- c. void
- d. mutator

8. When the value of an item is dependent on other data, and that item is not updated when the other data is changed, what has the value become?

- a. bitter
- b. stale
- c. asynchronous
- d. moldy

9. This is a method that is automatically called when an instance of a class is created.

- a. accessor
- b. constructor
- c. void
- d. mutator

10. When a local variable has the same name as a field, the local variable's name does this to the field's name.

- a. shadows
- b. complements
- c. deletes
- d. merges with

11. This is automatically provided for a class if you do not write one yourself.

- a. accessor method
- b. default instance
- c. default constructor
- d. variable declaration

12. Two or more methods in a class may have the same name, as long as this is different.

- a. their return values
- b. their access specifier
- c. their parameter lists
- d. their memory address

13. The process of matching a method call with the correct method is known as _____.

- a. matching
- b. binding
- c. linking
- d. connecting

14. A class's responsibilities are _____.

- a. the objects created from the class
- b. things the class knows
- c. actions the class performs
- d. both b and c

15. True or False: The new operator creates an instance of a class. True

16. True or False: Each instance of a class has its own set of instance fields. True

17. True or False: When you write a constructor for a class, it still has the default constructor that Java automatically provides. False

18. True or False: A class may not have more than one constructor. False

19. True or False: To find the classes needed for an object-oriented application, you identify all of the verbs in a description of the problem domain. False

Find the Error

1. Find the error in the following class:

```
public class MyClass {  
    private int x;  
    private double y;  
    public void MyClass(int a, double b) {  
        x = a;  
        y = b;  
    }  
}
```

Constructor لا يمكن أن يكون له نوع إرجاع (void). يجب حذف void.

2. Assume the following method is a member of a class. Find the error:

```
public void total(int value1, value2, value3) {  
    return value1 + value2 + value3;  
}
```

خطآن:

- value2, value3 missing data types.
 - method return type void but returns value.
- التصحيح: `public int total(int value1, int value2, int value3)`

3. The following statement attempts to create a Rectangle object. Find the error:

```
Rectangle box = new Rectangle;
```

الإجابة:

Missing parentheses.

التصحيح: `Rectangle box = new Rectangle();`

4. Find the error in the following class:

```
public class TwoValues {  
    private int x, y;  
    public TwoValues() { x = 0; }  
    public TwoValues() { x = 0; y = 0; }  
}
```

constructor لا يمكن وجود أكثر من Duplicate constructors (same parameter list).
بنفس التوقيع.

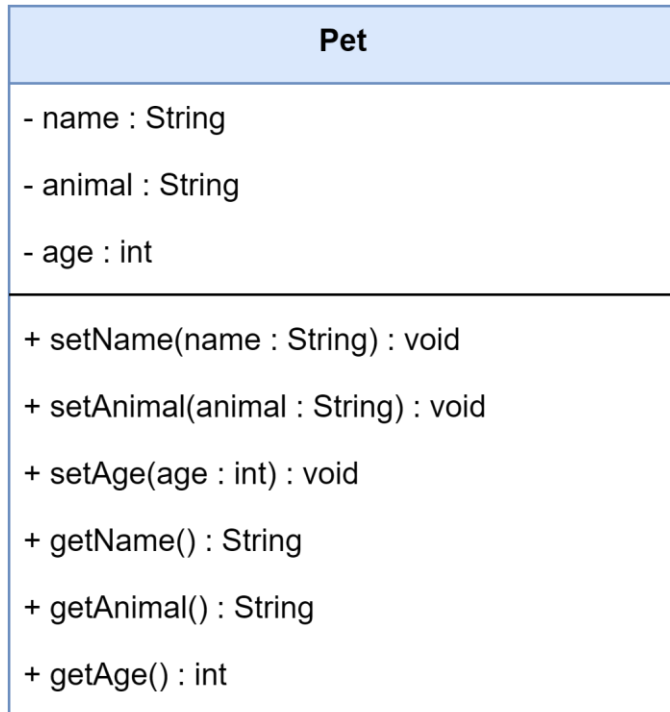
5. Find the error in the following class:

```
public class FindTheError {  
    public int square(int number) {  
        return number * number;  
    }  
    public double square(int number) {  
        return number * number;  
    }  
}
```

Overloading لا يتم بناءً على نوع الإرجاع فقط. يجب تغيير قائمة المعاملات.

Algorithm Workbench

1a. UML diagram for Pet class:



1b. Java code for Pet class:

```
public class Pet {  
    private String name;  
    private String animal;  
    private int age;  
  
    public void setName(String name) { this.name = name; }  
    public void setAnimal(String animal) { this.animal = animal; }  
    public void setAge(int age) { this.age = age; }  
    public String getName() { return name; }  
    public String getAnimal() { return animal; }  
    public int getAge() { return age; }  
}
```

2a. Constructor for Book class:

```
public Book(String title, String author, String publisher, int copiesSold)
{
    this.title = title;
    this.author = author;
    this.publisher = publisher;
    this.copiesSold = copiesSold;
}
```

2b. Accessor and mutator methods for Book:

```
public String getTitle() { return title; }
public void setTitle(String title) { this.title = title; }
public String getAuthor() { return author; }
public void setAuthor(String author) { this.author = author; }
public String getPublisher() { return publisher; }
public void setPublisher(String publisher) { this.publisher = publisher; }
public int getCopiesSold() { return copiesSold; }
public void setCopiesSold(int copiesSold) { this.copiesSold = copiesSold; }
```

3a. No-arg constructor for Square:

```
public Square() {
    sideLength = 0.0;
}
```

3b. Overloaded constructor for Square:

```
public Square(double sideLength) {
    this.sideLength = sideLength;
}
```

4a – Potential classes:

SavingsAccount, CheckingAccount, MoneyMarketAccount, Customer, Account

4b – Necessary classes:

Account

4c – Responsibilities of Account:

- balance (thing it knows)
- interest rate (thing it knows)
- deposit() (action)
- withdraw() (action)
- calculateInterest() (action)